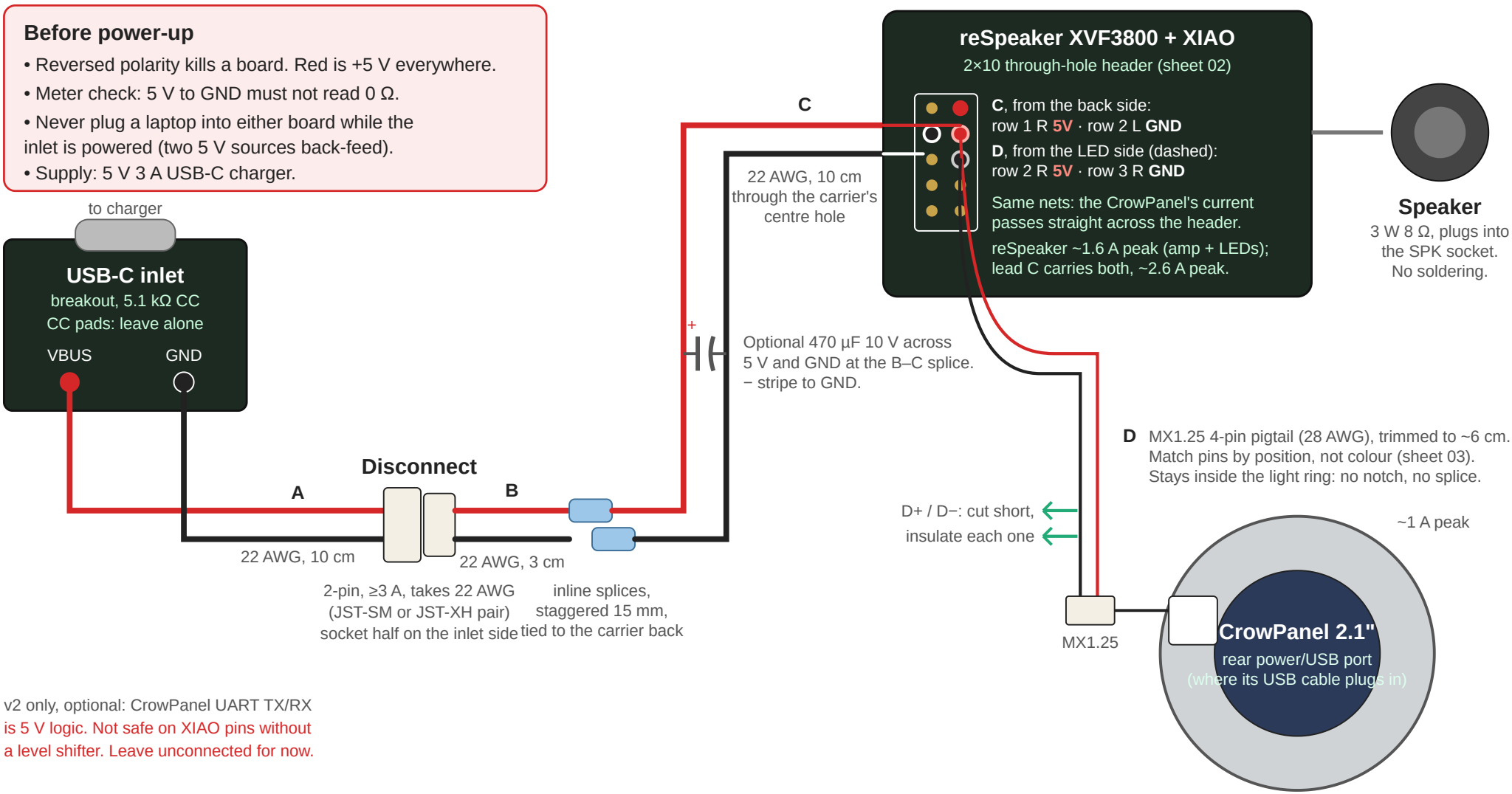
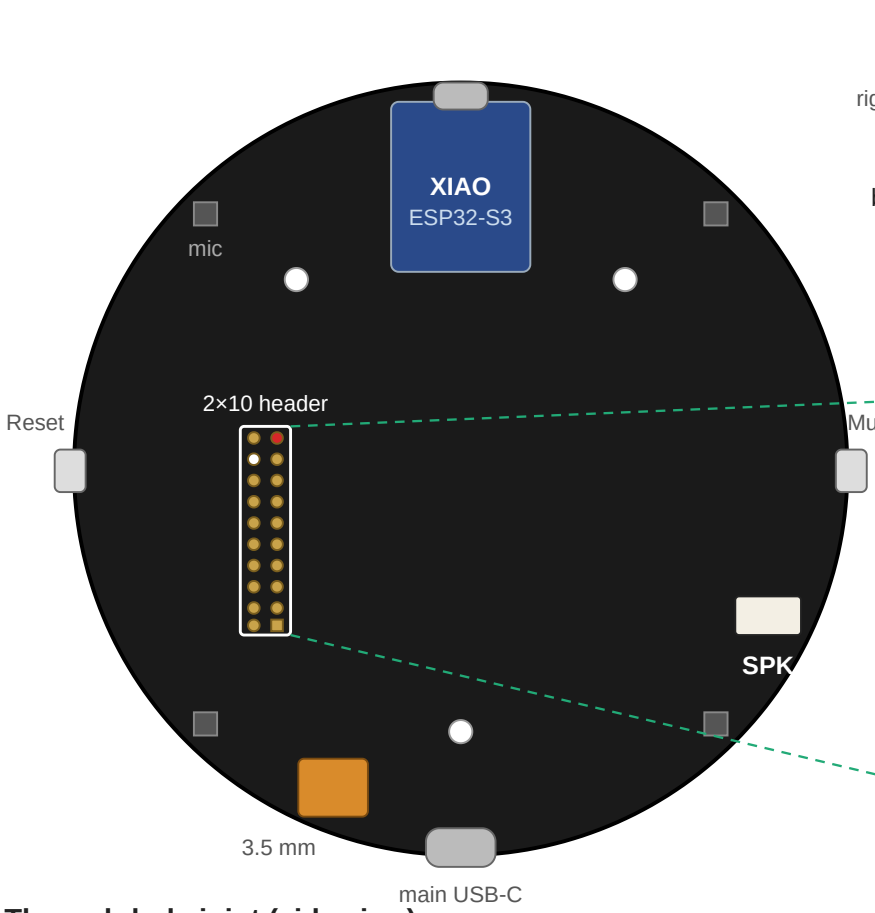
Voice Clock: power wiring

One 5 V USB-C inlet feeds the reSpeaker; the CrowPanel is fed from the reSpeaker's spare 5V / GND header pins. Red = +5 V, black = GND. Cut lengths; cut long and trim at the test fit. Not to scale.



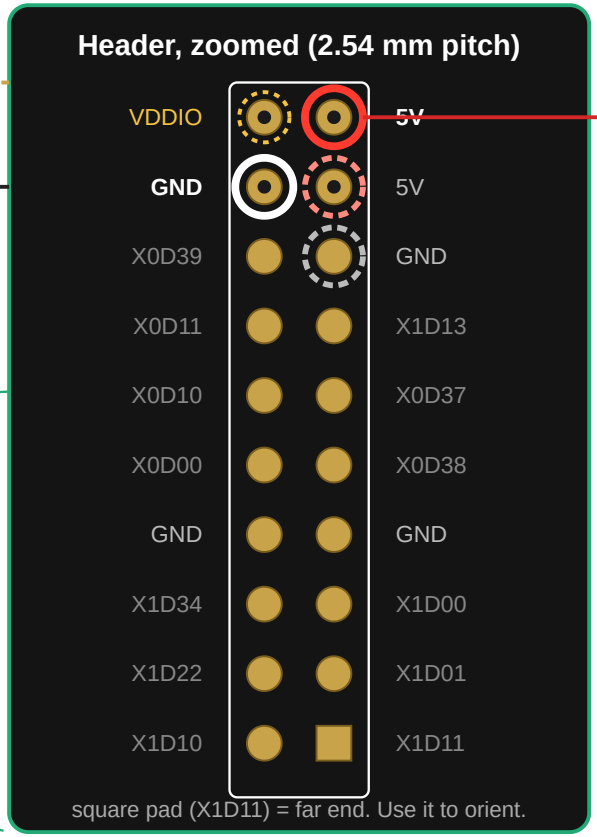
02 · reSpeaker XVF3800: 5 V and GND solder points (power in + CrowPanel out)

Back (component) side, XIAO at 12 o'clock, as in Seeed's pinout photo. Positions are approximate. Board Ø100 mm. Pin labels are from Seeed's pinout.



VDDIO ≠ 5 V
right next to +5 V.
Don't bridge it.

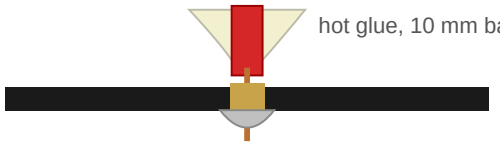
GND
black 22 AWG
row 2, left



+5 V
red 22 AWG
row 1, right

Lead D (CrowPanel)
dashed: row 2 right = 5V,
row 3 right = GND.
Pigtail wires go in from the
LED side, soldered on the back.

Through-hole joint (side view)



hot glue, 10 mm back from the joint

back side (faces the carrier)

LED side: 5 mm gap to the spider, keep it flat

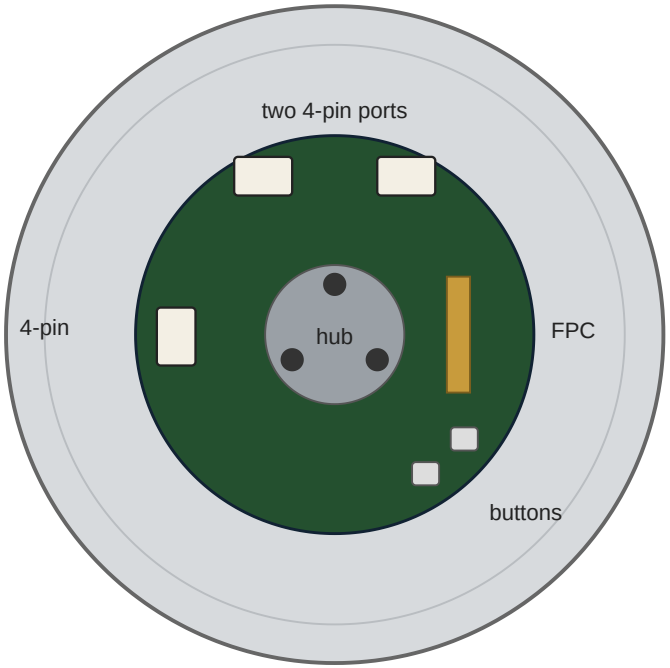
Strip 2–3 mm, tin, push in from the back, solder on the LED side, trim to ≤ 1 mm.

Do this first (2 min, bench)

1. Power the reSpeaker from its own USB-C. Meter header 5V to GND: ~5.0 V.
2. Unplug. Beep header 5V to the XIAO 5V pin: should beep (same rail).
3. GND holes sit on a ground plane: 370 °C, a chisel tip, hold 3–4 s.
4. If pins are already fitted, solder to the pin stub and heat-shrink it.

03 · CrowPanel 2.1": power through a plug-in pigtail

No soldering on the CrowPanel itself. Feed 5 V into the same 4-pin MX1.25 port that its supplied USB cable uses, with a spare pigtail wired to the reSpeaker header, so the panel stays removable.



Rear view (layout approximate)

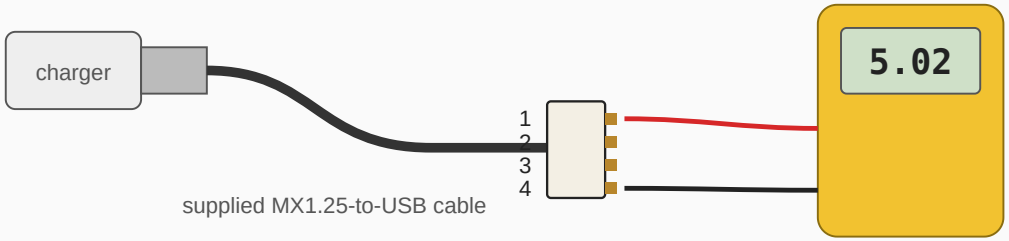
Use the port its MX1.25-to-USB cable plugs into: the one you flashed it through. Mark it with a dot.

Why not solder to the board?

Elecrow's schematic routes 5 V into the main board only through its 12-pin FPC (pins 1–2 = 5 V, 11–12 = GND) and a tiny test pad. A plug-in lead avoids heat on the display and keeps the panel removable.

Its UART pins are level-shifted to 5 V logic (v2 note).

A · Find 5 V and GND on the supplied cable (meter on DC V)



Probe pairs of contacts with a needle on each probe. **+5.0 V**: red probe = 5 V, black = GND.

Write down the two positions before you go on. Pigtail wire colours are not a standard; positions are.

B · Spare MX1.25 4-pin pigtail = lead D, straight into the reSpeaker header



Into the header:

Strip 2 mm, tin, slide 1.5 mm shrink on. Push each wire into its hole from the LED side and solder on the back side (the side facing the carrier). Trim flat on the LED side: only the bare pigtail crosses the 5 mm gap to the spider.

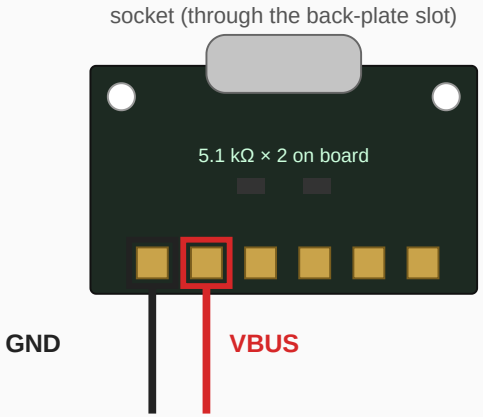
C · Test before the case

Plug the pigtail into the marked port and feed 5 V from the harness (or a bench supply set to 5.0 V, 1 A limit). The clock face should boot. If nothing happens, unplug at once and re-check step A.

04 · USB-C inlet, disconnect and B–C splice

Lead A stays with the back plate. Everything after the disconnect stays with the board stack, so the back plate can come off.

USB-C breakout (22 × 13 mm)

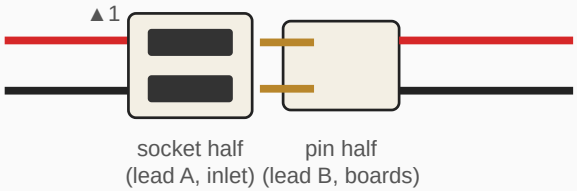


Pad order and names vary (VBUS may read V, VCC or 5V; GND may read G). Go by your board's silkscreen.

Leave CC1 / CC2 and D+ / D− alone.

Check: CC1→GND and CC2→GND each ≈ 5.1 kΩ.

Disconnect (2-pin, wire-to-wire)



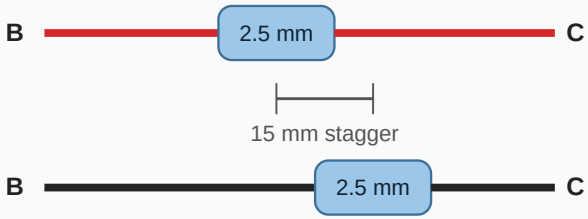
The live (inlet) side gets the shrouded sockets, so a loose lead can't short. Pin 1 = red.

Which connector

- ✓ JST-SM 2-pin (LED-strip pigtails): 3 A, 22 AWG
- ✓ JST-XH 2-pin male/female pair: 3 A, 22–28 AWG
- ✗ JST-PH: 2 A, crimps take 24–32 AWG only

Buy pre-wired pairs, then solder 22 AWG onto their leads.

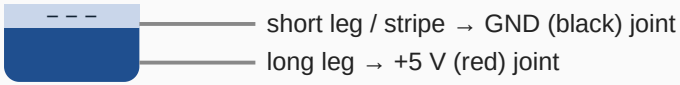
B–C splice (one per colour)



B: the disconnect pin half's own leads (22 AWG).
C: 22 AWG on to the reSpeaker header, through the carrier's centre hole. Slide the sleeve on first. Strip 6 mm, lay side by side, twist, solder, let cool, shrink.

There is no Y any more: the CrowPanel lead (D) starts at the reSpeaker header's spare 5V / GND (sheet 02).

Optional 470 µF / 10 V capacitor



Solder its legs into the two B–C joints before shrinking. Keep the legs short and sleeved. It smooths the amp's current peaks.

Iron and materials

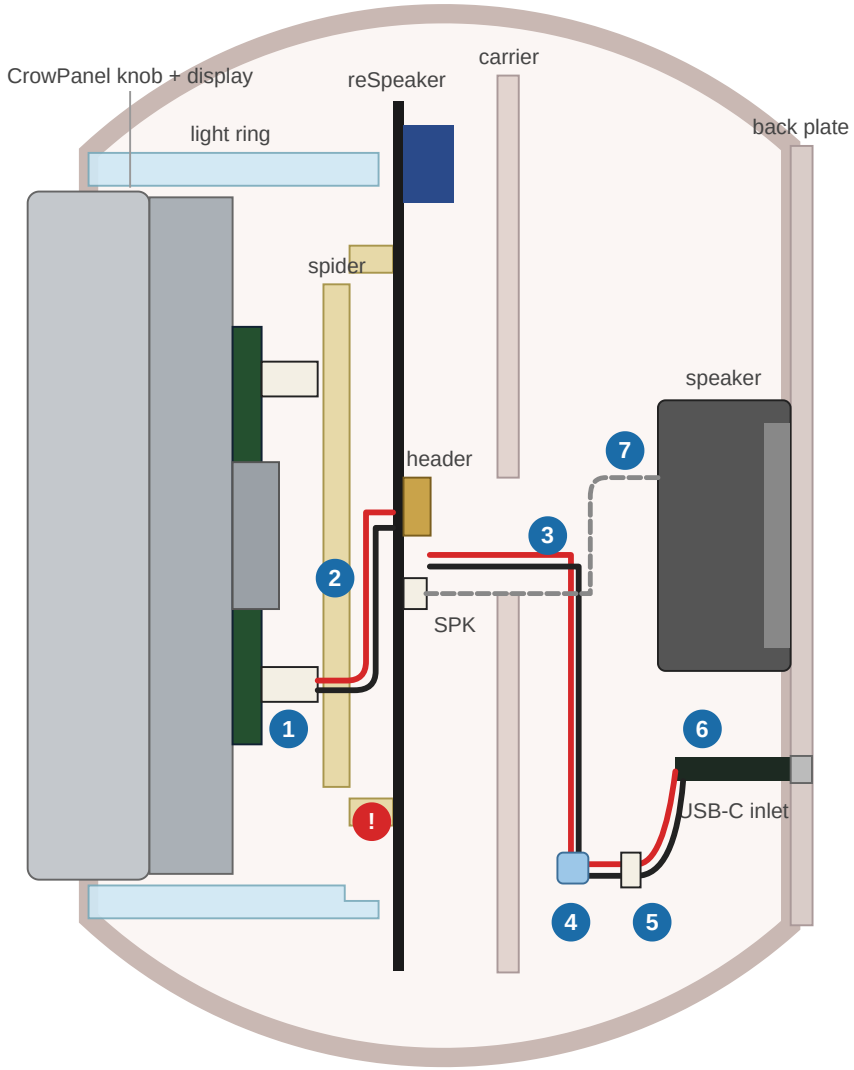
- Swap the heat-set tip for a 2 mm chisel. 340 °C for wire, 370 °C for ground-plane pads.
- 0.6–0.8 mm flux-core solder (63/37 is easiest), extra flux pen for the header.
- Silicone wire: 22 AWG red + black. Heat shrink 1.5, 2.5, 4.8 mm.
- Tin each stripped end before joining. Let joints cool before you move them.

Order of work

1. Bench checks: sheet 02 (reSpeaker) and sheet 03 step A.
2. Lead A: solder to the breakout, then to the disconnect socket half.
3. Lead B: the pin half's own leads, trimmed to ~3 cm.
4. Lead C: 22 AWG to the reSpeaker header (sheet 02).
5. Lead D: pigtail into the header's spare pins (sheet 03 B).
6. Thread C through the carrier hole, then splice B–C (+ cap).
7. Meter: 5 V to GND not 0 Ω at the disconnect pins.
8. Charger into the inlet, meter at the socket half: +5.0 V, red probe on pin 1. Then plug together: both boards boot.
9. Play audio loud: header 5V–GND stays above 4.75 V.
10. Hot-glue or tie the B–C splice to the carrier back.

05 · Where the wires run (side section)

Front (knob) on the left, back plate on the right. Parts to scale (4.5 px/mm) from the CAD, face tilt left out. Wires drawn schematically.



1. Pigtail (lead D) plug in the CrowPanel power port.
2. Between the spider arms and across the 5 mm gap to the header's spare 5V / GND holes, soldered from the LED side. All of it inside the light ring: no notch, nothing at the edge.
3. Lead C through the carrier's centre hole to the header's 5V / GND from the back side.
4. B–C inline splices (+ optional capacitor). Hot-glue or zip-tie them to the carrier back so no tug reaches a joint.
5. Disconnect. Unplug it to lift the back plate off. Clear of the speaker and the inlet cage.
6. Lead A to the USB-C inlet. Stays with the back plate. 10 cm gives room to set the plate beside the clock.
7. Speaker lead (plug-in, no soldering) through the centre hole to the SPK socket. Zip-tie it to its anchor.

! **No splices, heat shrink or loops in the 5 mm gap** between the spider and the reSpeaker. Only the bare pigtail passes there; trim its joints flat on the LED side.

The carrier's centre hole is the only wire opening in the stack. Nothing runs round the board edge, so the finished stack drops into the rear shell at its keyed angle with every lead plugged in, and the front shell twist-locks on with no wires attached to it.

Leave about 2 cm of slack on lead C behind the carrier.